

TruHealth

Report

Christopher Hughett

Age: 37 | Sex: Male

ID#: N49KZCE4ZM494952

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CLIA Lab Director: Melissa Keinath, PhD FACMG

TruDiagnostic

881 Corporate Drive

Lexington, KY 40503

Welcome to your personalized report on Epigenetic Biomarkers

What are Epigenetic Biomarkers?

Epigenetic biomarkers are specific biological markers derived from analyzing DNA methylation patterns in your genome. Unlike traditional blood-based metrics of your health such as serum levels, which provide a single snapshot in time, epigenetic biomarkers uniquely offer insights into ongoing biological functions by predicting levels of nutrients, metabolites, and other essential biochemical entities. These epigenetic measurements are based on patterns in your DNA methylation landscape, influenced by both genetic and environmental factors. Epigenetic biomarkers provide a more stable and comprehensive view of your health over time, reflecting long-term exposures and trends rather than short-term fluctuations.

Epigenetic Biomarkers vs Serum Levels

Serum levels represent a direct measurement of substances in your blood at a specific point in time. These levels can vary based on recent meals, exercise, stress, and other transient factors. In contrast, epigenetic biomarkers are derived from DNA methylation patterns, providing a broader and more consistent indication of health, which isn't as influenced by short-term changes. This makes epigenetic biomarkers especially useful for understanding long-term health trends and risks.

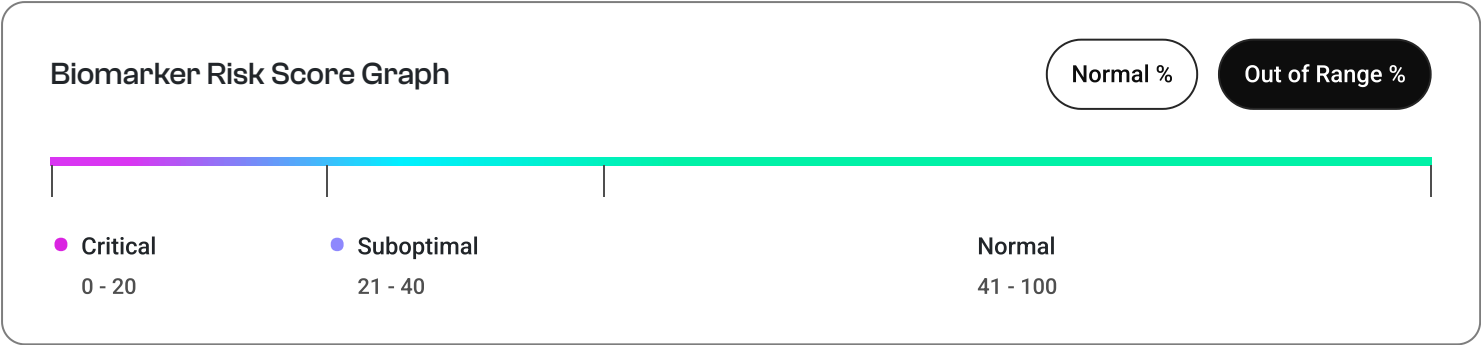
Epigenetic biomarkers provide a more stable and comprehensive view of your health over time, reflecting long-term exposures and trends rather than short-term fluctuations.

The Reference Cohort: The Data Behind TruHealth's Analytics

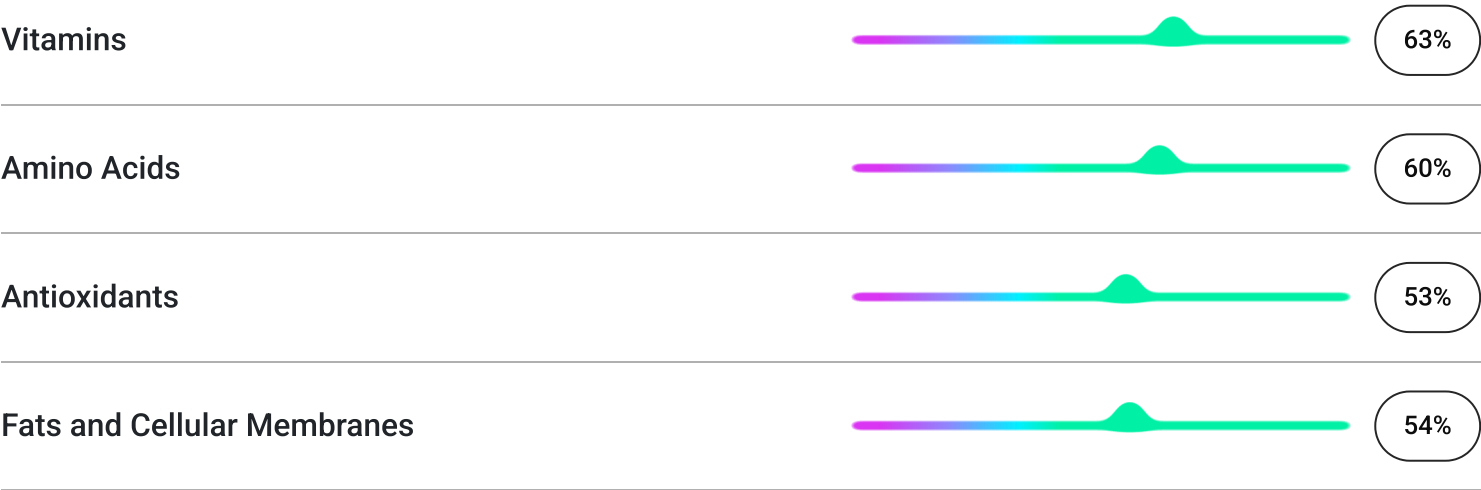
Our algorithms and biomarkers are developed and validated using data from a subset of the Massachusetts General Brigham (MGB) Biobank. The MGB Biobank is a comprehensive repository containing over 130,000 high-quality samples from more than 100,000 consented patients. These samples are linked to detailed EMR data, which includes the patients' lifetime medical histories. The biobank also incorporates survey data on lifestyle, environment, and family history. To ensure meaningful percentile comparisons in your report, we used an equal combination of data from the MGB Biobank and TruDiagnostic patients, providing a robust, diverse, and extensively studied reference population.

Biomarkers Overview

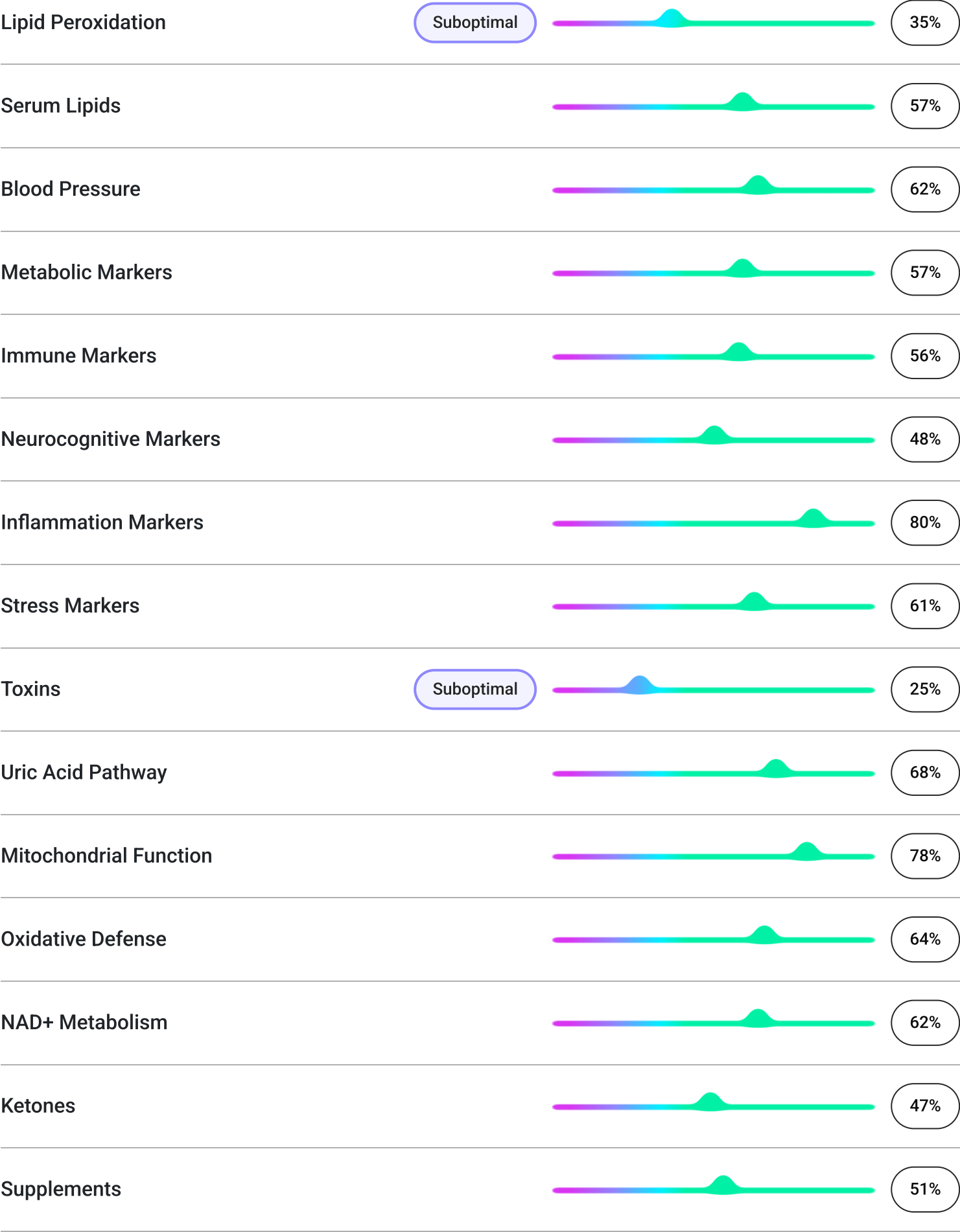
This section of the TruHealth report provides a comprehensive overview of an individual's general health and nutrition markers, each derived from one or more epigenetic biomarkers that reflect underlying cellular health and function. These markers are presented in a normalized percentile range, allowing for a straightforward comparison with a reference population. A score of 0 indicates the worst outcome, representing potential areas of concern or risk, while a score of 100 signifies the best possible outcome, highlighting areas of optimal health. The percentile ranking provides an easy-to-interpret indication of how an individual's health status compares to the broader population, offering valuable insights into their relative standing in terms of metabolic health, immune function, inflammation, mitochondrial activity, and more. This comparison against a carefully studied cohort gives context to these markers, enabling users to understand areas where they may need intervention or where they are already performing well.



NUTRITION



GENERAL HEALTH MARKERS



TOP PERSONALIZED RECOMMENDATIONS

FATS AND CELLULAR MEMBRANES

SFA

Increase saturated fatty acids (SFA) by adding moderate amounts of coconut oil, grass-fed butter, full-fat dairy, and fatty meats to your diet while maintaining overall balance with healthy fats.

🔗 MCT oil

🔗 C15

VITAMINS

Vitamin B8 (Inositol)

Increase inositol through foods like fruits, grains, nuts, and avoid excess caffeine.

🔗 Inositol

NEUROCOGNITIVE MARKERS

Dopamine Metabolites

Exercise regularly, sleep well, manage stress, and eat tyrosine-rich foods.

🔗 Tyrosine

🔗 Mucuna Pruriens

AMINO ACIDS

L-Aspartic Acid

High-intensity interval training (HIIT).

🔗 N-acetylcysteine (NAC)

🔗 L-Theanine

NAD+ METABOLISM

Nicotinamide riboside (NR, Precursor)

Increase intake of B3-rich foods, exercise, and intermittent fasting to naturally boost NR and NAD+.

🔗 Nicotinamide Riboside (NR)

🔗 Nicotinamide Mononucleotide (NMN)

TOXINS

Polycyclic aromatic hydrocarbons (Air Pollution)

Avoid environmental PAH exposure (pollution, smoking). Increase detox pathways.

🔗 Sulforaphane

🔗 Alpha-Lipoic Acid

EPIGENETIC BIOMARKERS RISK SCORES

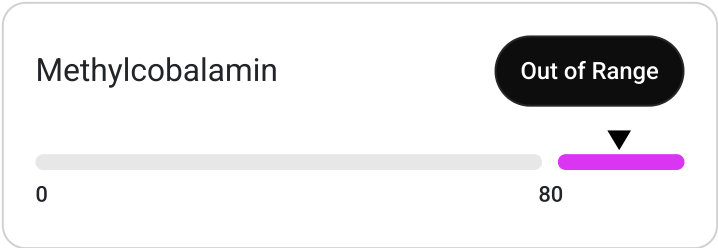
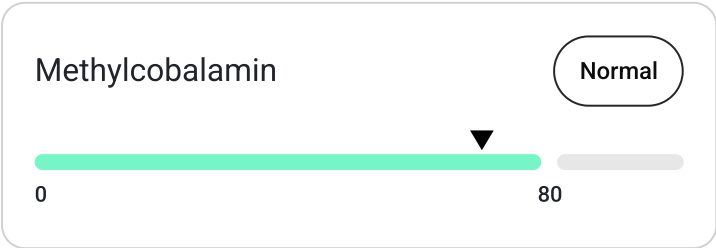
The **Epigenetic Biomarkers Risk Scores** section of the report provides a detailed breakdown of specific **epigenetic biomarkers** derived from your epigenome, each linked to relevant **risk scores** and associated **biomarkers**. Every epigenetic biomarker is accompanied by a concise explanation of its function in the body and its relationship to disease or health outcomes. The **optimal and suboptimal ranges** for these biomarkers vary, depending on whether the biomarker is most beneficial at **high, low, or midrange levels**. For biomarkers identified as suboptimal, personalized **lifestyle** and **supplement recommendations** are provided for improvement. Additionally, prior epigenetic biomarker values are displayed to **track changes over time**, offering insight into progress and areas that may need continued attention.

There are Three Different Ranges for Each Epigenetic Biomarkers (EB)

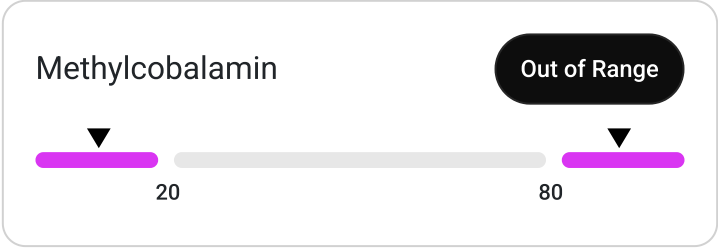
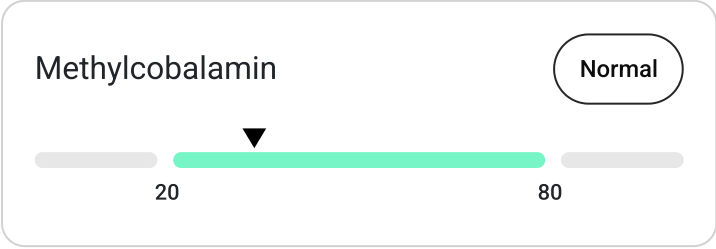
In the below section, each epigenetic biomarker will be presented as your percentile when compared against the reference population. Each epigenetic biomarker is identified by our science team as ideal when LOW, MIDDLE, or HIGH. When the ideal biomarker is thought to be LOW, a percentile of 0% is best. When the MIDDLE is ideal, 50% is best. When HIGH is ideal, 100% is best.

Normal Out of Range ▲ Critically Out of Range

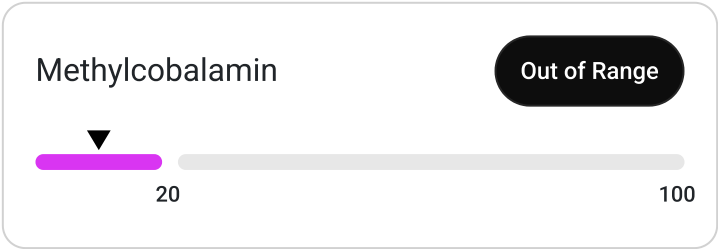
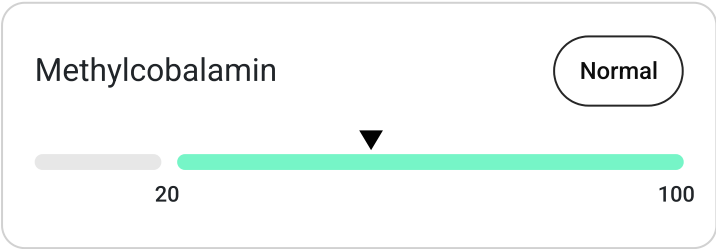
Example: Normal Range is **LOW**



Example: Normal Range is **MIDDLE**



Example: Normal Range is **HIGH**



VITAMINS

Vitamin A_m

Retinol (vitamin A)_m



Description

Retinol (Vitamin A) is essential for vision, immune function, and cellular communication.

Vitamin B2 (Riboflavin)_m

Riboflavin (vitamin B2)_m



Description

A B-vitamin essential for energy production and cellular function

Vitamin B3 (Nicotinamide)_m

Nicotinamide_m



Description

Nicotinamide is a form of vitamin B3, essential for NAD⁺ production and cellular energy. Low levels are associated with lower NAD⁺ production, and high levels are associated with inhibition of Sirtuins and NAD⁺ recycling pathways.

Vitamin B5 (Pantothenic Acid)_m

Pantothenic Acid (vitamin B5)_m



Description

Pantothenic Acid (Vitamin B5) is essential for CoA synthesis and energy metabolism.

Vitamin B6 (P5P)_m

Pyridoxine, Pyridoxine-5-Phosphate (vitamin B6)_m



Description

Pyridoxine (Vitamin B6) is a coenzyme in amino acid metabolism and neurotransmitter synthesis.

Vitamin B8 (Inositol)_m

Inositol (myoinositol)_m

Low



3%

Description

Inositol is a carbohydrate involved in cell membrane formation and insulin signal transduction.

Recommendations

Increase inositol through foods like fruits, grains, nuts, and avoid excess caffeine.

Inositol

Vitamin C_m

Ascorbic acid 2-sulfate_m



18%

Description

Sulfated form of Vitamin C.

Vitamin C_m

Ascorbic acid 3-sulfate_m



33%

Description

Another sulfated form of Vitamin C.

Vitamin D_m

Cholecalciferol (vitamin D)_m



76%

Description

A form of Vitamin D; important for bone health and immune function.

Vitamin E_m

Alpha-tocopherol_m



82%

Description

A form of Vitamin E with strong antioxidant properties. High levels can be associated with reduced gamma-tocopherol function, and low levels can be associated with poor antioxidant function.

Choline_m

Total cholines_m



86%

Description

Essential nutrient involved in brain health and fat metabolism.

Betaine_m

5

100

41%

Description

A methyl donor involved in liver function and cell hydration.

AMINO ACIDS

Methionine_m

5

95

91%

Description

An essential amino acid involved in protein synthesis and detoxification processes.

Cysteine_m

5

95

19%

Description

Amino acid involved in protein synthesis and antioxidant functions

S-methylmethionine_m

5

100

77%

Description

A methylated form of methionine involved in methylation reactions

Taurine_m

5

100

95%

Description

A sulfur-containing amino acid with antioxidant properties.

Ergothioneine_m

5

100

56%

Description

Ergothioneine is a powerful antioxidant involved in cellular protection and oxidative stress regulation. Low levels could indicate reduced cellular defense against oxidative damage.

Glutamine_m 79%

An amino acid essential for gut health.

Arginine_m 100%

An amino acid involved in protein synthesis and nitric oxide production.

L-Aspartic Acid_m

High

5 90

98%

L-Aspartic Acid is a non-essential amino acid important in the urea cycle and energy production. Low levels may impair protein synthesis and neurotransmitter function, while high levels could indicate metabolic stress.

High-intensity interval training (HIIT).

 GABA

Valine_m
N-carbamoylvaline_m

5 90 60%

N-carbamoylvaline_m

N-carbamoylvaline is a derivative of valine, playing a role in nitrogen metabolism. Abnormal levels might indicate stress on nitrogen metabolism from diet or inflammation.

Asparagine_m 79%

An amino acid involved in metabolic processes.

Threonine_m

A horizontal bar chart for Threonine_m. The bar is green and extends from 5 to 80 on a scale. A black triangle points down to the bar at the 23% mark. The value 23% is displayed in a circle on the right.

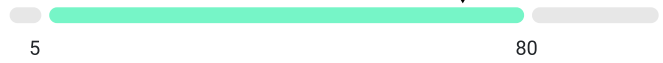
N-acetylthreonine_m

5 80

23%

N-acetylthreonine_m

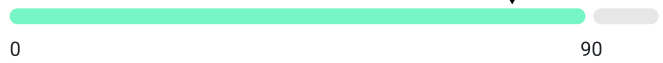
N-acetylthreonine is an acetylated derivative of threonine, essential for protein production. Low levels could impair immune function and recovery, while high levels might suggest dietary imbalances.

N-acetylvaline_m

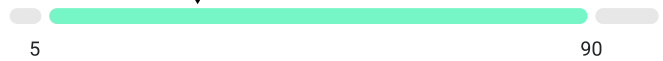
N-acetylvaline is involved in valine metabolism, an essential amino acid. Low levels could affect muscle recovery, while high levels could reflect metabolic stress.



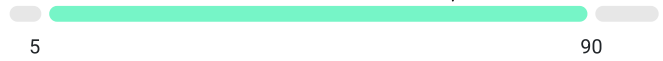
An amino acid involved in detox and sleep.

N-acetylcarnosine_m

A derivative of carnosine, which acts as an antioxidant in the body.



Intermediate in methionine metabolism and cysteine biosynthesis

N-acetylhistidine_m

A derivative of histidine, important for metal ion binding and antioxidant functions. Low levels could affect muscle recovery, while high levels may indicate inflammation or stress.



Non-essential amino acid involved in nitric oxide production and vascular health.

Carotenoids_m

Lutein_m

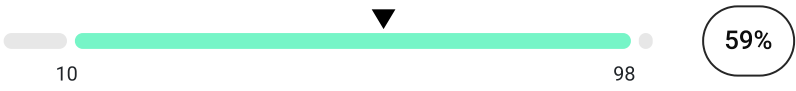


Description

A carotenoid with antioxidant properties, found in green leafy vegetables.

Carotenoids_m

Carotene diol_m



Description

Carotenoid found in plants; antioxidant properties.

Acetyl-L-Carnitine_m



Description

Acetyl-L-Carnitine is a compound involved in fatty acid metabolism and mitochondrial energy production. Low levels could suggest impaired mitochondrial function or reduced fatty acid utilization.

Vitamin A_m

Retinol (vitamin A)_m



Description

Retinol (Vitamin A) is essential for vision, immune function, and cellular communication.

Vitamin E_m

Alpha-tocopherol_m



Description

A form of Vitamin E with strong antioxidant properties.

Ergothioneine_m



Description

Ergothioneine is a powerful antioxidant involved in cellular protection and oxidative stress regulation. Low levels could indicate reduced cellular defense against oxidative damage.

Vitamin B2 (Riboflavin)_m

Riboflavin (vitamin B2)_m



Description

A B-vitamin essential for energy production and cellular function

Ascorbic acid 2-sulfate_m

A sulfated form of Vitamin C.

Ascorbic acid 3-sulfate_m



Another sulfated form of Vitamin C.

 Ω_{-3_m} 

Omega-3 is a polyunsaturated fatty acid with essential roles in reducing inflammation and supporting heart, brain, and joint health. High levels may indicate adequate dietary intake, while low levels could suggest poor dietary consumption or increased risk of inflammatory conditions.

Docosahexaenoate (DHA; 22:6n3)_m

An omega-3 fatty acid that plays a key role in brain health and inflammation.

Docosapentaenoic acid_m

A lesser-known omega-3 fatty acid involved in inflammatory processes.

Eicosapentaenoate (EPA; 20:5n3)_m

An omega-3 fatty acid with anti-inflammatory effects, found in fish oils.

Omega 6_m

Omega-6_m



Description

Omega-6 is a polyunsaturated fatty acid that supports cell membrane integrity and helps regulate inflammatory processes. Elevated levels may reflect a diet rich in whole-food sources (e.g., nuts, seeds, plant oils) and have been linked with lower all-cause and cardiovascular mortality. Low levels could indicate inadequate intake of essential fats or dysregulated lipid metabolism.

LA_m



Description

Linoleic Acid (LA) is an essential omega-6 polyunsaturated fatty acid critical for cell membrane integrity and inflammatory processes. Excessive LA intake may promote lipid peroxidation and chronic inflammation, potentially negatively affecting metabolic health and longevity.

PUFA_m



Description

Polyunsaturated Fatty Acids (PUFAs) are essential fats involved in cell structure and inflammation regulation. Excessive omega-6 PUFA intake may promote lipid peroxidation, leading to oxidative stress and inflammation, which could negatively impact longevity and metabolic health.

MUFA_m



Description

Monounsaturated Fatty Acids (MUFAs) are healthy fats with a single double bond, involved in maintaining cell membrane fluidity and supporting cardiovascular health. MUFAs may reduce inflammation and oxidative stress, potentially benefiting metabolic health and longevity.

SFA_m



Description

Saturated Fatty Acids (SFAs) are fats with no double bonds, commonly found in animal products and some plant oils. High intake of SFAs can increase cholesterol levels and may promote inflammation, potentially impacting cardiovascular health and longevity when consumed in excess.

Recommendations

Increase saturated fatty acids (SFA) by adding moderate amounts of coconut oil, grass-fed butter, full-fat dairy, and fatty meats to your diet while maintaining overall balance with healthy fats.

-  MCT oil
-  C15

Pentadecanoate_m

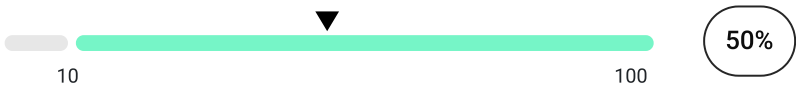
Pentadecanoate (C15:0)_m



Description

Pentadecanoate (C15:0) is a saturated fatty acid with potential anti-inflammatory properties. High levels may indicate metabolic stress or excessive intake of specific dietary fats, while low levels could suggest impaired fat metabolism.

Phosphoglycerides_m



Description

Phosphoglycerides are a class of phospholipids that form a major component of cell membranes, supporting membrane fluidity, signaling, and energy metabolism. Imbalances in phosphoglycerides may disrupt cellular function, potentially affecting metabolic health and longevity.

Phosphatidylcholine_m

Phosphatidylcholines_m



Description

Phosphatidylcholines are a major class of phospholipids critical for cell membrane structure, lipid transport, and signaling. They support liver function and cognitive health, with imbalances potentially affecting metabolism and longevity.

Sphingomyelins_m



Description

Sphingomyelins are a type of sphingolipid essential for cell membrane integrity and signaling, particularly in nerve and brain tissues. Dysregulation of sphingomyelins may contribute to metabolic dysfunction and neurodegenerative diseases, potentially impacting longevity.

LIPID PEROXIDATION

Phospholipase A2_m

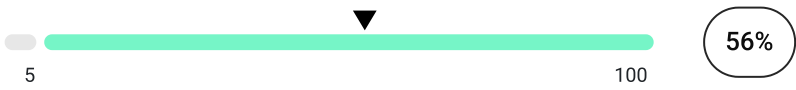


Description

Phospholipase A2 is an enzyme involved in lipid metabolism and inflammatory regulation. High levels may indicate heightened inflammatory activity or cellular damage, while low levels could suggest impaired lipid signaling or reduced inflammatory response.

Glutathione peroxidase_m

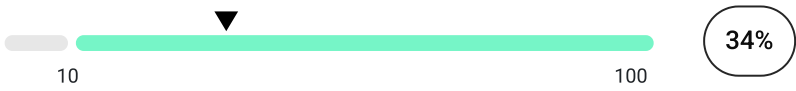
Glutathione peroxidase 1 (GPX1)_m



Description

Glutathione peroxidase is an enzyme involved in antioxidant defense and cellular protection. Elevated levels may reflect increased oxidative stress or an adaptive response to cellular damage, while low levels could indicate impaired antioxidant capacity and increased vulnerability to oxidative damage.

Octadecadienedioate (C18:2-DC)_m



Description

C18:2-DC is a dicarboxylic acid, linked to lipid metabolism and energy regulation. High levels could be linked to metabolic disorders, while low levels may indicate impaired lipid metabolism or energy production.

SERUM LIPIDS

ApoB_m



Description

Apolipoprotein B (ApoB) is a structural protein of atherogenic lipoproteins, including LDL and VLDL. Elevated ApoB levels reflect the number of these lipoproteins, providing a stronger predictor of cardiovascular disease risk and atherosclerosis than LDL cholesterol alone, making it crucial for assessing metabolic health and longevity.

LDL-C_m



Description

LDL-C (Low-Density Lipoprotein Cholesterol) represents the cholesterol carried by LDL particles, often referred to as "bad cholesterol." Elevated LDL-C levels are strongly associated with increased risk of atherosclerosis and cardiovascular disease, making it a key marker for assessing cardiovascular health and longevity.

LDL particle size_m



Description

Smaller, dense LDL particles are more atherogenic and strongly associated with cardiovascular disease risk, while larger LDL particles are less likely to penetrate arterial walls, indicating a potentially lower risk profile and better metabolic health.



Description

VLDL-C (Very-Low-Density Lipoprotein Cholesterol) represents cholesterol carried by VLDL particles, which transport triglycerides from the liver to peripheral tissues. Elevated VLDL-C is associated with increased atherogenic risk, contributing to triglyceride-rich lipoprotein accumulation, cardiovascular disease, and metabolic dysfunction, negatively impacting longevity.



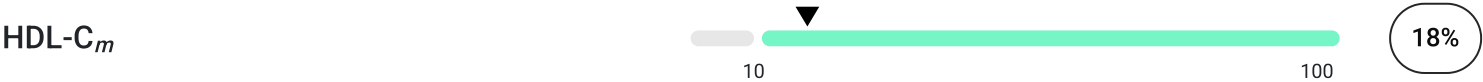
Description

Smaller VLDL particles are more atherogenic and linked to higher cardiovascular risk, while larger particles may indicate improved lipid metabolism and reduced metabolic dysfunction, influencing overall health and longevity.



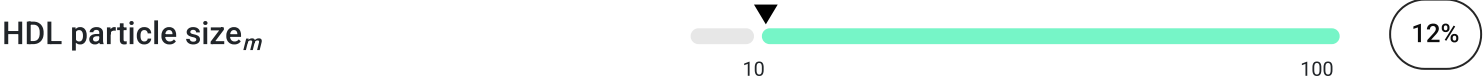
Description

Apolipoprotein A1 (ApoA1) is the primary structural protein of HDL particles, playing a key role in reverse cholesterol transport and antioxidant defense. Higher ApoA1 levels are associated with improved cardiovascular health, reduced atherosclerosis risk, and better longevity outcomes through enhanced lipid metabolism and anti-inflammatory effects.



Description

HDL-C (High-Density Lipoprotein Cholesterol) represents the cholesterol carried by HDL particles, often referred to as "good cholesterol." Higher HDL-C levels are associated with improved reverse cholesterol transport, reduced cardiovascular disease risk, and better longevity outcomes through anti-inflammatory and antioxidant effects.



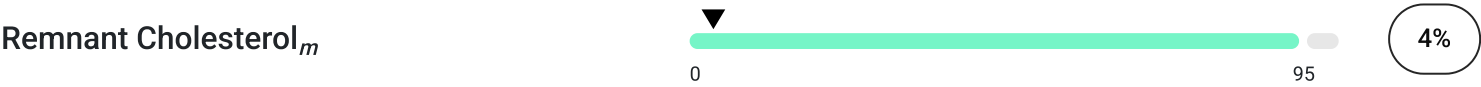
Description

Larger HDL particles are generally more effective at reverse cholesterol transport and provide stronger antioxidant and anti-inflammatory benefits, contributing to reduced cardiovascular disease risk and improved longevity.



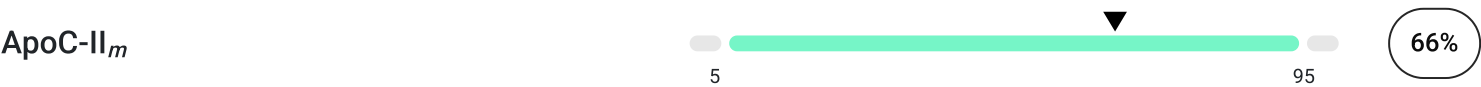
Description

Total triglycerides are the primary form of fat in the blood, stored for energy use in adipose tissue. Elevated levels are associated with insulin resistance, metabolic syndrome, and increased cardiovascular risk, making them a critical marker for assessing metabolic health and longevity.



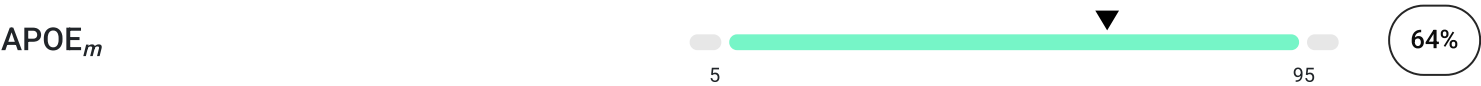
Description

Cholesterol carried in triglyceride-rich particles. Higher levels associated with causal ASCVD risk and pancreatitis risk when TG are very high. Lower levels associated with lower residual risk.



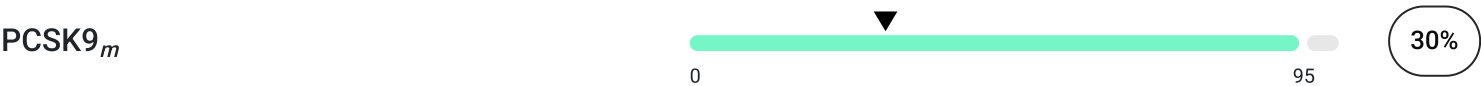
Description

Protein that helps clear triglycerides from the blood; imbalance can track with high triglycerides and heart risk. Higher levels associated with high triglycerides. Lower levels associated with rare genetic deficiency and chylomicronemia.



Description

Protein that moves fats and cholesterol; certain forms increase risk for Alzheimer’s and heart disease. Higher levels associated with remnant phenotype and dyslipidemia. Lower levels associated with impaired trafficking. Caveat: does not measure ApoE subtype - requires genetic testing.



Description

Regulates LDL (‘bad’) cholesterol levels; higher activity means higher cholesterol and more artery risk. Lower levels associated with lower risk. Caveat: PCSK9 inhibitors lower serum levels and statins can raise serum levels, epigenetic signal may be altered with these meds.

BLOOD PRESSURE

Vanillactic Acid (VAA)_m

Vanillactic Acid_m

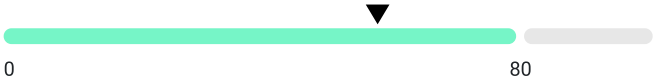


43%

Description

Vanillactic Acid (VAA) is a metabolite derived from phenylalanine metabolism. Elevated levels of VAA are associated with increased cardiovascular disease (CVD) risk and hypertension (HTN) and may indicate pathways contributing to vascular inflammation, oxidative stress, or hormonal regulation of blood pressure. Conversely, low levels of VAA might reflect insufficient phenylalanine metabolism or altered gut microbiome activity, which could indicate disruptions in metabolic or microbial processes.

Phenylacetylglutamine_m



58%

Description

Phenylacetylglutamine (PAG) is a gut-derived metabolite linked to phenylalanine metabolism. Elevated PAG levels are associated with increased cardiovascular and kidney disease risk, systemic inflammation, and reduced longevity.

Dimethylarginine, SDMA and ADMA_m



44%

Description

Markers that block blood vessel relaxation; higher levels are linked to kidney disease and heart risk. Higher levels associated with cardiovascular events and reduced GFR. Lower levels associated with better Nitric Oxide bioavailability and renal handling. Caveat: ACEi, ARB, SGLT2 inhibitors, PDE-5 inhibitors, and statins can change serum faster than epigenetic signal.

METABOLIC MARKERS

HgbA1c_m



64%

Description

Measure of blood sugar levels over a period of time, indicative of glucose control.

Glucose_m



3%

Description

Blood sugar level, related to diabetes risk.

Fat Burning Marker_m

Palmitoylcarnitine_m



Description

Palmitoylcarnitine plays a role in fatty acid transport into mitochondria for oxidation. High levels can signal impaired fatty acid oxidation, while low levels might affect energy production from fats.

Satiety Hormone_m

Leptin_m

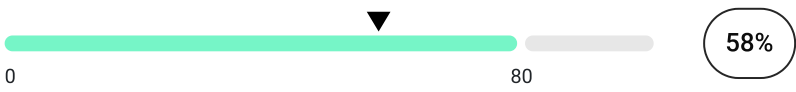


Description

Leptin is a hormone produced by fat cells that helps regulate appetite and energy balance by signaling the brain to reduce hunger and increase energy expenditure. High levels may indicate leptin resistance and excess body fat, while low levels can suggest insufficient fat reserves or impaired metabolic function.

Phenylalanine Dysbiosis Marker_m

Phenylacetylglutamine_m



Description

Phenylacetylglutamine (PAG) is a gut-derived metabolite linked to phenylalanine metabolism. Elevated PAG levels are associated with increased cardiovascular and kidney disease risk, systemic inflammation, and reduced longevity.

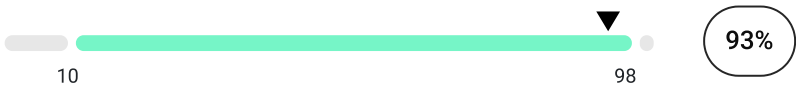
IGF-1_m



Description

Growth factor that supports muscle and metabolism. Lower levels associated with frailty and sarcopenia. Higher levels associated with cancer risk and acromegaly or GH therapy. Caveat: GH therapy or anabolic steroids can elevate serum while the epigenetic marker may remain lower.

Adiponectin_m



Description

Hormone from fat tissue that improves insulin sensitivity. Lower levels associated with insulin resistance and visceral adiposity. Very high levels associated with heart failure or catabolic states in older adults.

Cystatin-C_m



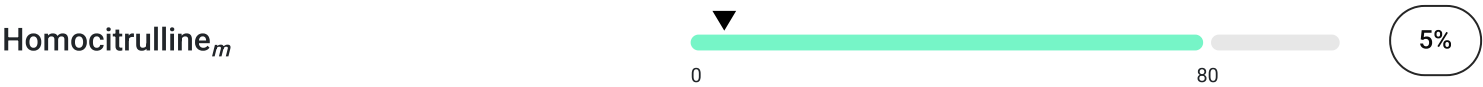
Description

Kidney function measure, not muscle independent like creatinine. Higher levels associated with chronic kidney disease, cardiovascular events, and all-cause mortality. Lower levels associated with preserved filtration.



Description

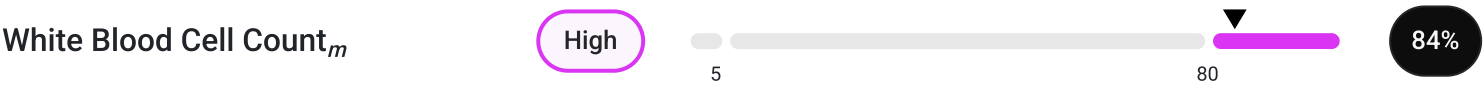
Sugar alcohol made in the body and found in sugar-free foods. Higher levels associated with frequent sugar-free intake and cardiometabolic risk signals in observational studies. Lower levels reflect minimal exposure.



Description

Marker of protein damage from nitrogen waste, often elevated in kidney disease and tied to heart and vessel risk. Higher levels associated with CKD or uremia, atherosclerosis, heart failure, and mortality in CKD cohorts. Lower levels reflect lower carbamylation burden.

IMMUNE MARKERS



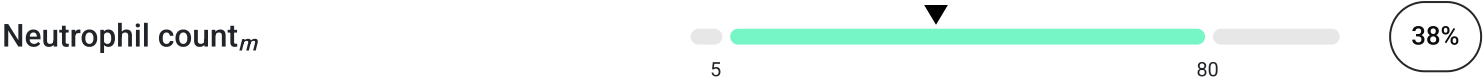
Description

Measures the amount of white blood cells in the blood, important for immune function.

Recommendations

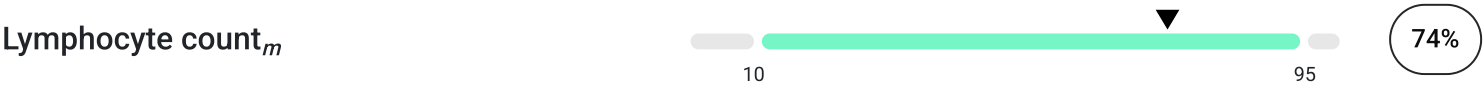
Address immune disorders, chronic inflammation or infections.

- Omega-3s
- Curcumin
- Quercetin
- Vitamin D
- Green Tea Extract



Description

Neutrophil count indicates levels of neutrophils, important for immune defense.



Description

A type of white blood cell, important for immune system function.

Inflammation marker, linked to cardiovascular and chronic diseases

A horizontal bar chart with a green bar representing the 'Yes' response. The bar starts at 5 on the x-axis and ends at 95. A black triangle points to the bar. To the right of the bar, a circle contains the text '26%'.

Response	Percentage
Yes	26%

Neutrophil-to-lymphocyte ratio (NLR) is a marker of systemic inflammation and immune balance. High ratios are associated with chronic inflammation, cardiovascular disease, and poor immune regulation, while low ratios may indicate impaired immune response or hematological abnormalities.

Systemic Immune-Inflammation Index (SII) is a composite marker calculated as (Platelet count × Neutrophil count) / Lymphocyte count. High levels are linked to systemic inflammation, poor prognosis in cancer, and cardiovascular risk, while low levels may indicate immune suppression or reduced inflammatory response.

Low

30 80

24%

The CD4/CD8 ratio reflects the balance between helper and cytotoxic T cells, indicating immune system status. High ratios are linked to autoimmune disorders or chronic inflammation, while low ratios suggest immune suppression, aging, or increased risk of infections and certain cancers.

Boost a low CD4/CD8 ratio by improving sleep, reducing stress, eating a nutrient-dense diet, and addressing potential infections or immune deficiencies with medical guidance.

 Zinc

NEUROCOGNITIVE MARKERS

Memory Health Protein_m

Neurogranin_m



38%

Description

Neurogranin is a calmodulin-binding protein, linked to synaptic plasticity and cognition. High levels can be linked to neurodegenerative diseases (e.g., Alzheimer's), while low levels might impair synaptic plasticity and cognitive function.

Brain Inflammation Marker_m

Quinolate_m



15%

Description

Quinolate is a metabolite in the kynurenine pathway linked to neuroinflammation. High levels are linked to neurotoxicity and neurodegenerative disorders, while low levels may impair tryptophan metabolism and affect immune function.

Dopamine Metabolites_m

Dopamine 3-O-sulfate_m

Low ▲



1%

Description

Dopamine 3-O-sulfate is a dopamine metabolite involved in neurotransmitter regulation and detoxification. Elevated levels may indicate altered dopamine metabolism, while low levels could suggest reduced detoxification or neurotransmitter imbalance.

Recommendations

Exercise regularly, sleep well, manage stress, and eat tyrosine-rich foods.

- Tyrosine
- Mucuna Pruriens
- SAMe
- Vitamin B6
- Magnesium

Cell Repair Marker_m

Transforming growth factor beta (TGF-beta)_m



13%

Description

Transforming Growth Factor Beta (TGF-beta) is a cytokine involved in cell growth, proliferation, differentiation, and apoptosis, with key roles in immune regulation and inflammation. Elevated levels may indicate heightened immune activity or chronic inflammation, while low levels could suggest impaired cellular signaling or immune response.

Brain Anti-inflammatory Protein_m

Progranulin_m



39%

Description

Progranulin is a protein involved in wound healing, inflammation, and neurodegeneration. Elevated levels may reflect active inflammation or tissue repair, while low levels could indicate impaired healing processes or an increased risk of neurodegenerative conditions.

VGFM

Neurosecretory protein VGF_m

5

95

72%

Description

Brain-derived protein that supports mood, memory, and energy balance. Lower levels associated with cognitive decline and depression. Higher levels associated with exercise adaptation. Caveat: may be altered by antidepressants.

INFLAMMATION MARKERS

IL-6_m

0

80

62%

Description

IL-6 (Interleukin-6) is a cytokine involved in inflammation and the immune response.

Oxidative Stress Marker_m

Methionine sulfone_m

0

80

21%

Description

An oxidized form of methionine, a marker of oxidative stress. High levels may indicate excessive oxidative stress or impaired sulfur amino acid metabolism, while low levels could affect protein synthesis and antioxidant production.

Serum amyloid A-1 protein_m

0

90

1%

Description

Serum Amyloid A-1 Protein is an acute-phase protein involved in inflammation and immune response. Elevated levels may indicate active inflammation, while low levels could reflect a lack of inflammatory activity.

Glycoprotein acetyls_m

0

90

9%

Description

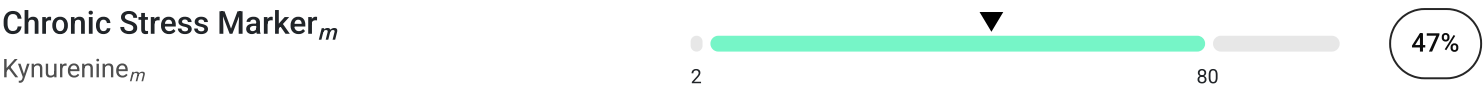
Glycoprotein acetyls (GlycA) are inflammatory biomarkers representing glycosylated acute-phase proteins in the bloodstream. Elevated GlycA levels are associated with systemic inflammation, increased cardiovascular disease risk, and poorer metabolic health, making it a key marker for assessing longevity and chronic disease burden.



Description

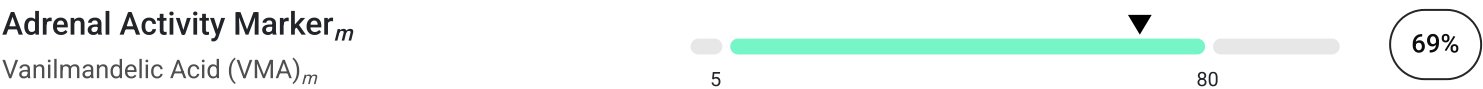
Inflammation marker, linked to cardiovascular and chronic diseases

STRESS MARKERS



Description

Kynurenine is a metabolite of tryptophan, involved in immune modulation and neurobiology. High levels are associated with inflammation and neurodegenerative diseases, while low levels may indicate impaired tryptophan metabolism.



Description

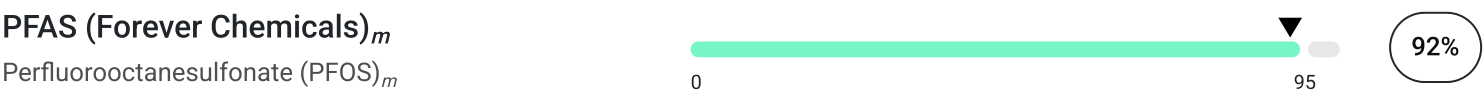
VMA is a metabolite of catecholamines (epinephrine and norepinephrine), and high levels are linked to catecholamine excess, while low levels might indicate adrenal insufficiency or low catecholamine turnover.



Description

Long-term adrenal androgen availability. Lower levels associated with frailty, depressed mood, and higher mortality in older adults. Higher levels associated with PCOS or supplementation. Caveat: exogenous DHEA, testosterone, estrogen, or progesterone can make the epigenetic signal appear low despite normal or high serum.

TOXINS



Description

PFOS is a synthetic chemical found in industrial and consumer products, including nonstick cookware, water-repellent fabrics, and stain-resistant treatments.

PFAS (Forever Chemicals)_m

Perfluorooctanoate (PFOA)_m

0

95

74%

Description

Perfluorooctanoate (PFOA) is a synthetic chemical used in the production of nonstick cookware, waterproof fabrics, and certain industrial processes.

Acrolein (Pollution and Smoking)_m

S-(3-hydroxypropyl)mercapturic acid (HPMA)_m

0

95

46%

Description

HPMA is a metabolite involved in the detoxification of chemicals like acrolein, found in cigarette smoke, exhaust fumes, and industrial emissions.

Polycyclic aromatic hydrocarbons (Air Pollution)_m

2-hydroxyfluorene sulfate_m

High

0

95

98%

Description

2-Hydroxyfluorene sulfate is a metabolite that indicates exposure to polycyclic aromatic hydrocarbons (PAHs), which are found in tobacco smoke, grilled or charred foods, and industrial emissions.

Recommendations

Avoid environmental PAH exposure (pollution, smoking). Increase detox pathways.

- Sulforaphane

Alpha-Lipoic Acid

N-acetylcysteine (NAC)

Glutathione

Pesticides_m

Glyphosate_m

0

80

60%

Description

Glyphosate is a broad-spectrum herbicide commonly used to control weeds, known for its role in agricultural practices but controversial due to potential health and environmental risks.

Lead Exposure_m

Bone Lead Predictor_m

0

95

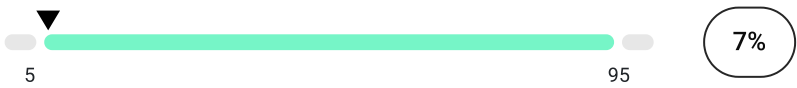
81%

Description

This measure reflects long-term lead accumulation in the body, particularly within the skeleton, which stores lead for decades. Elevated levels can be associated with chronic health risks, including hypertension and cognitive decline, due to lead's prolonged presence in bone tissue and periodic release into the bloodstream.

Uric Acid_m

Urate_m



Description

Uric acid is an antioxidant byproduct of purine metabolism. High levels may indicate risk for gout, CVD, HTN, and metabolic disorders, while low levels could reduce antioxidant defense and increase oxidative stress.

Xanthine_m



Description

Xanthine is a purine metabolite involved in nucleotide breakdown. Elevated levels may indicate oxidative stress or impaired purine metabolism, potentially contributing to inflammation and reduced longevity.

Allantoin_m



Description

Allantoin is a purine metabolite formed from the oxidation of uric acid. Elevated levels may indicate oxidative stress, potentially linked to inflammation, tissue damage, and reduced longevity. In humans, it serves as a marker of increased ROS activity rather than a precursor to uric acid

MITOCHONDRIAL FUNCTION

Energy Balance Marker_m

Dynamin-1-like protein (DRP1)_m



Description

DRP1 is a protein involved in mitochondrial fission, crucial for cellular energy production and regulation of apoptosis. Elevated levels are linked to mitochondrial fission and could contribute to mitochondrial dysfunction

ATP synthase (Enzyme)_m

ATP synthase subunit beta (ATP5B)_m



Description

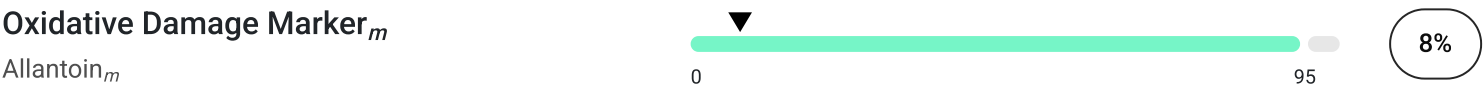
A component of the mitochondrial ATP synthase complex, high levels are associated with cellular stress and mitochondrial dysfunction.

OXIDATIVE DEFENSE



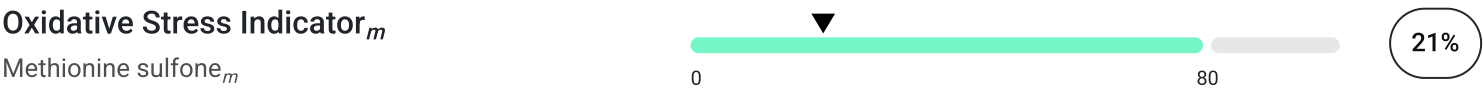
Description

An enzyme released during inflammation and used as a marker for oxidative stress.



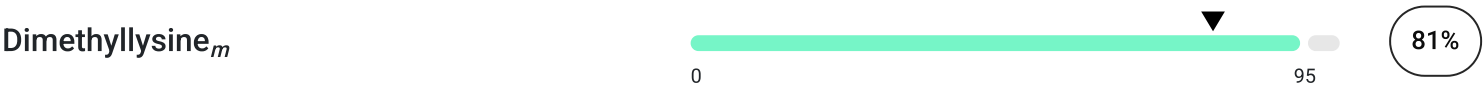
Description

Allantoin is formed through oxidation of uric acid and thus serves as a marker of oxidative stress. Higher levels of allantoin in the blood increased oxidative stress, inflammation, or other stress-related metabolic disturbances.



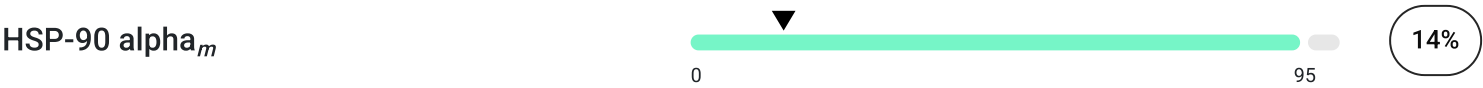
Description

An oxidized form of methionine, a marker of oxidative stress. High levels may indicate excessive oxidative stress or impaired sulfur amino acid metabolism, while low levels could affect protein synthesis and antioxidant production.



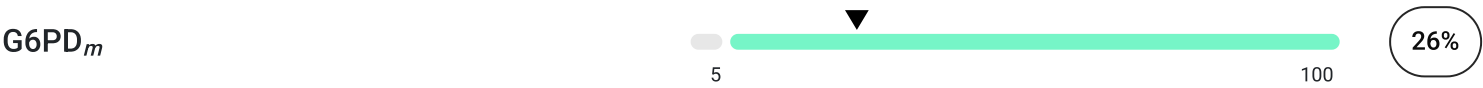
Description

Strain on protein repair and folding systems. Higher levels associated with inflammation, insulin resistance, CKD, and higher cardiometabolic risk. Lower levels reflect lower cumulative damage.



Description

"Chaperone" protein that helps other proteins fold. Higher levels associated with cancer progression, inflammatory states, and adverse prognosis. Lower levels associated with lower chaperone demand.



Description

Enzyme that protects red blood cells from stress. Lower levels associated with inherited G6PD deficiency and hemolysis risk. Higher levels generally benign.

NAD+ METABOLISM

Byproduct marker (indicates inefficiency in NAD recycling)_m

N1-methyl-2-pyridone-5-carboxamide (2PY)_m



Description

N1-methyl-2-pyridone-5-carboxamide (2PY) is a metabolite of NAD+ that can accumulate with increased NAD+ production. Elevated 2PY levels may inhibit enzymes like PARP, potentially impairing DNA repair and cellular energy metabolism, which could negatively affect longevity.

Nicotinamide riboside (NR, Precursor)_m

Nicotinamide riboside_m



Description

Nicotinamide Riboside is a precursor to NAD+. Low levels can lead to reduced NAD+ production and decreased cellular energy, while high levels of NR indicate efficient NAD+ synthesis but could suggest over-reliance on supplementation.

Recommendations

Increase intake of B3-rich foods, exercise, and intermittent fasting to naturally boost NR and NAD+.

-  Nicotinamide Riboside (NR)
-  Nicotinamide Mononucleotide (NMN)
-  Vitamin B3 (Niacin)

Nicotinamide (Precursor)_m

Nicotinamide_m



Description

Nicotinamide is a form of vitamin B3, essential for NAD+ production and cellular energy. Low levels are associated with lower NAD+ production, and high levels are associated with inhibition of Sirtuins and NAD+ recycling pathways.

KETONES

Beta Hydroxybutyrate_m

3-hydroxybutyrate (BHBA)_m



Description

3-Hydroxybutyrate (BHBA) is a ketone body produced during fat metabolism, primarily in the liver, and serves as an alternative energy source for the brain and muscles, especially during periods of low carbohydrate intake or fasting.

SUPPLEMENTS

Alpha-ketoglutarate_m



22%

Description

Alpha-ketoglutarate is a key intermediate in the Krebs cycle, crucial for energy production, amino acid metabolism, and cellular function regulation.

Spermidine_m



59%

Description

Spermidine is a polyamine compound involved in cellular growth, proliferation, and apoptosis, known for its potential role in promoting autophagy and longevity, and is found in foods like aged cheese, soy products, and whole grains.